

DAX PRACTICE #3

Time Intelligence & Advanced Measures

- SUMX & VAR patterns
- YoY, MoM, YTD, QTD, MTD
- Power BI Desktop

Total Revenue 2

```
1 Total Revenue 2 =  
2 SUMX(  
3     Fact_Sales,  
4     Fact_Sales[OrderQuantity] * RELATED(Product_Lookup[ProductPrice])  
5 )
```

What it does:

- SUMX is an iterator - goes row by row through Fact_Sales
- For each row, multiplies OrderQuantity by ProductPrice (pulled in via RELATED)
- After every row is calculated, adds all the results together

Output: One total revenue number - the sum of every row's calculated revenue, based on whatever filter is applied (year, product, region, etc.)

Total Profit 2

```
1 Total Profit 2 =  
2 VAR _Total_Revenue =  
3 SUMX(  
4     Fact_Sales,  
5     Fact_Sales[OrderQuantity] * RELATED(Product_Lookup[ProductPrice])  
6 )  
7  
8 VAR _Total_COGS =  
9 SUMX(  
10    Fact_Sales,  
11    Fact_Sales[OrderQuantity] * RELATED(Product_Lookup[ProductCost])  
12 )  
13  
14 RETURN  
15 _Total_Revenue - _Total_COGS
```

What it does:

- VAR calculates revenue and cost once, then stores them as short, reusable names
- Then subtracts cost from revenue to get profit

Output: One total profit number - built from clear, reusable steps instead of one long formula.

Profit Margin 2

```
1 Profit Margin 2 =  
2 VAR _Total_Revenue =  
3 SUMX(  
4     Fact_Sales,  
5     Fact_Sales[OrderQuantity] * RELATED(Product_Lookup[ProductPrice])  
6 )  
7  
8 VAR _Total_COGS =  
9 SUMX(  
10     Fact_Sales,  
11     Fact_Sales[OrderQuantity] * RELATED(Product_Lookup[ProductCost])  
12 )  
13  
14 VAR _Total_Profit = _Total_Revenue - _Total_COGS  
15  
16 RETURN  
17 DIVIDE(_Total_Profit,_Total_Revenue,0)
```

What it does:

- Uses VAR to calculate revenue, cost, and profit first
- Divides profit by revenue using DIVIDE, which returns blank instead of an error if revenue is 0

Output: Profit margin shown as a percentage, safe from divide-by-zero errors.

COGS Ratio

```
1 COGS Ratio =  
2 VAR _Total_Revenue =  
3 SUMX(  
4     Fact_Sales,  
5     Fact_Sales[OrderQuantity] * RELATED(Product_Lookup[ProductPrice])  
6 )  
7  
8 VAR _Total_COGS =  
9 SUMX(  
10    Fact_Sales,  
11    Fact_Sales[OrderQuantity] * RELATED(Product_Lookup[ProductCost])  
12 )  
13  
14 RETURN  
15 DIVIDE(_Total_COGS,_Total_Revenue,0)
```

What it does:

- Same VAR + DIVIDE pattern as profit margin
- Divides cost by revenue instead of profit by revenue

Output: Cost of goods sold (COGS) shown as a percentage of revenue.

Previous Year Sales - DATEADD

```
1 Previous Year Sales =  
2 CALCULATE(  
3     [Total Revenue],  
4     DATEADD('Calendar-Table'[Date], -1, YEAR)  
5 )
```

What it does:

- CALCULATE changes the filter, then recalculates revenue
- DATEADD shifts every date back by 1 year (Number = -1, Interval = YEAR)

Output: Revenue from the same period last year.

Previous Year Sales - SAMEPERIODLASTYEAR

```
1 Previous Year Sales =  
2 CALCULATE(  
3     [Total Revenue],  
4     SAMEPERIODLASTYEAR('Calendar-Table'[Date])  
5 )
```

What it does:

- Returns the same dates as this year, just shifted back one year
- Same result as DATEADD(-1, YEAR), just a shorter, simpler function name

Output: Same result as the DATEADD version - just a cleaner, shorter formula.

Year Wise Sales Diff

$$1 \text{ Year Wise Sales Diff} = [\text{Total Revenue}] - [\text{Previous Year Sales}]$$

What it does:

- Takes Current Year Sales
- Subtracts Previous Year Sales: Current Year Sales - Previous Year Sales

Output: How much revenue went up or down compared to last year.

YoY Sales Growth %

```
1 YoY Sales Growth % =  
2 DIVIDE([Year Wise Sales Diff],[Previous Year Sales],0)
```

What it does:

- Divides the yearly difference by Previous Year Sales
- DIVIDE keeps it safe from errors when last year's revenue is 0

Output: Growth compared to last year, shown as a percentage.

YoY Sales Growth % 2

```
1 YoY Sales Growth % 2 =  
2 VAR _PY_Sales =  
3 CALCULATE([Total Revenue],DATEADD('Calendar-Table'[Date],-1,YEAR))  
4  
5 VAR _YoY_Diff = [Total Revenue]-_PY_Sales  
6  
7 RETURN  
8 DIVIDE(_YoY_Diff,_PY_Sales,0)
```

What it does:

- Same YoY growth logic, written as a single measure
- VAR stores current year, previous year, and the difference, then reuses them

Output: Same growth % as before - confirms the formula is correct.

YoY Output - Sliced by Year

Year

☐ 2020

☒ 2021

☐ 2022

Year	Month Name	Total Revenue	Previous Year Sales	Year Wise Sales Diff	YoY Sales Growth %	YoY Sales Growth % 2
2021	Jan	432,425.74	585,312.65	-152,886.91	-26.12%	-26.12%
2021	Feb	474,162.79	532,226.25	-58,063.46	-10.91%	-10.91%
2021	Mar	471,961.88	643,436.10	-171,474.23	-26.65%	-26.65%
2021	Apr	494,957.42	653,364.04	-158,406.62	-24.24%	-24.24%
2021	May	545,534.74	659,325.90	-113,791.15	-17.26%	-17.26%
2021	Jun	533,824.98	669,988.67	-136,163.69	-20.32%	-20.32%
2021	Jul	815,356.47	486,115.01	329,241.46	67.73%	67.73%
2021	Aug	804,193.39	536,452.82	267,740.57	49.91%	49.91%
2021	Sep	952,743.49	344,062.87	608,680.62	176.91%	176.91%
2021	Oct	1,029,821.05	404,276.60	625,544.45	154.73%	154.73%
2021	Nov	1,133,913.05	326,611.15	807,301.89	247.18%	247.18%
2021	Dec	1,635,308.80	563,761.53	1,071,547.27	190.07%	190.07%
Total		9,324,203.79	6,404,933.58	2,919,270.21	45.58%	45.58%

What it does:

- Table filtered to year 2021
- Shows revenue, last year's revenue, the difference, and both growth % formulas side by side

Output: Both growth % formulas match every month - proof the two versions calculate identically.

MoM Sales Growth % 2

```
1 MoM Sales Growth % 2 =  
2 VAR _PM_Sales =  
3 CALCULATE([Total Revenue],DATEADD('Calendar-Table'[Date],-1,MONTH))  
4  
5 VAR _MoM_Diff = [Total Revenue]-_PM_Sales  
6  
7 RETURN  
8 DIVIDE(_MoM_Diff,_PM_Sales,0)
```

What it does:

- Same idea as YoY, but DATEADD shifts back 1 month instead of 1 year
- MoM Sales Growth % = MoM Sales Different / Previous Month Sales

Output: Growth compared to last month, shown as a percentage.

YTD Sales

```
1 YTD Sales =  
2 CALCULATE(  
3     [Total Revenue],DATESYTD('Calendar-Table'[Date])  
4 )
```

What it does:

- DATESYTD returns every date from Jan 1 to the current date in the filter
- CALCULATE adds up revenue across those days - restarts fresh every new year

Output: Running total of sales so far this year.

YTD Output - Daily Running Total

Date	Total Revenue	YTD Sales
1/1/2020	8,351.46	8,351.46
1/2/2020	14,313.08	22,664.54
1/3/2020	28,041.32	50,705.86
1/4/2020	17,713.07	68,418.93
1/5/2020	7,855.64	76,274.56
1/6/2020	21,266.34	97,540.90
1/7/2020	8,554.74	106,095.64
1/8/2020	25,365.43	131,461.07
1/9/2020	14,313.08	145,774.15
1/10/2020	14,109.80	159,883.95
1/11/2020	31,619.59	191,503.54
1/12/2020	25,047.89	216,551.43
1/13/2020	7,855.64	224,407.07
1/14/2020	31,669.59	256,076.66
Total	24,914,586.82	9,185,449.45

What it does:

- Table shows revenue next to YTD sales for each day - YTD keeps accumulating as the year goes on

Output: By Jan 14, YTD sales reached 256,076.66 - the running total so far.

MTD Sales

```
1 MTD Sales =  
2 CALCULATE(  
3     [Total Revenue], DATESMTD('Calendar-Table'[Date])  
4 )
```

What it does:

- DATESMTD returns every date from the 1st of this month to today - restarts each new month

Output: Running total of sales so far this month.

QTD Sales

```
1 QTD Sales =  
2 CALCULATE(  
3     [Total Revenue], DATESQTD('Calendar-Table'[Date])  
4 )
```

What it does:

- DATESQTD returns every date from the start of this quarter to today - restarts each new quarter

Output: Running total of sales so far this quarter.

QTD Output Resets Each Quarter

Date	Total Revenue	QTD Sales
3/17/2020	25,732.97	1,466,757.29
3/18/2020	22,867.82	1,489,625.10
3/19/2020	11,433.91	1,501,059.01
3/20/2020	31,847.87	1,532,906.88
3/21/2020	14,313.08	1,547,219.96
3/22/2020	20,884.78	1,568,104.74
3/23/2020	17,688.07	1,585,792.81
3/24/2020	42,354.40	1,628,147.21
3/25/2020	17,688.07	1,645,835.28
3/26/2020	26,267.81	1,672,103.09
3/27/2020	10,734.81	1,682,837.90
3/28/2020	28,626.16	1,711,464.06
3/29/2020	21,266.34	1,732,730.40
3/30/2020	14,313.08	1,747,043.48
3/31/2020	13,931.52	1,760,975.00
4/1/2020	12,133.01	12,133.01
4/2/2020	10,556.53	22,689.54
4/3/2020	29,121.98	51,811.51
4/4/2020	24,641.33	76,452.84
4/5/2020	14,313.08	90,765.92
Total	24,914,586.82	5,123,233.36

What it does:

- Table tracks revenue and QTD sales through March
- QTD keeps accumulating through the quarter, reaching 1,760,975.00 by March 31

Output: On April 1, QTD resets to 12,133.01 - proof it starts over each new quarter.